

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

Project Title :
Tejas Nutrition Agent - AI Powered Personal Nutrition Assistant.
Domain of Project – Healthcare

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement -

Problem Statement : Nutrition Agent

The Challenge –

In an era where health awareness is growing, individuals increasingly seek personalized nutrition guidance. However, most existing tools provide generic diet plans, lack real-time adaptability, and fail to consider a person's holistic lifestyle, cultural preferences, allergies, and evolving health conditions. Furthermore, dietitians and nutritionists face limitations in scaling personalized consultations due to time and resource constraints.

Generative AI presents a groundbreaking opportunity to revolutionize this space by enabling an intelligent, interactive, and adaptive virtual nutrition assistant. By leveraging natural language processing (NLP), multimodal understanding, and large-scale dietary databases, an AI-powered assistant can generate dynamic meal plans, recommend smart food swaps, and explain nutritional choices—all tailored to the individual.

Many people struggle to maintain healthy eating habits because nutrition information is often scattered, confusing, and not personalized.

- Users frequently face problems such as:
- Difficulty creating balanced diet plans
- Lack of nutrition knowledge
- Different dietary preferences (vegetarian/non-vegetarian)
- Gym and fitness nutrition confusion
- Healthy meal planning
- Daily calorie and protein estimation
- As a result, people often fail to achieve their fitness or health goals.

Proposed Solution -

Proposed Solution: AI-Powered Nutrition Agent

The proposed solution is an AI-powered Nutrition Agent built using **IBM watsonx Orchestrate** and **IBM Granite Models**.

The proposed solution is **Tejas Nutrition Agent**, an AI-powered virtual nutrition assistant developed using IBM watsonx Orchestrate. The Nutrition Agent leverages IBM Granite large language models to understand user queries and provide scientifically informed, personalized dietary recommendations.

Unlike traditional rule-based chatbots, this intelligent agent understands conversational language and adapts responses according to the user's goals and preferences.

The agent acts as a virtual nutrition coach capable of:

- Answering nutrition-related questions
- Generating personalized diet plans
- Suggesting healthy meal routines
- Recommending post-workout meals
- Providing calorie and macronutrient guidance
- Giving healthy coffee optimization tips
- Encouraging healthy eating habits

The agent interacts naturally with users using conversational AI and delivers quick, personalized responses.

Technology Used

- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **IBM watsonx Orchestrate** - IBM watsonx Orchestrate is the primary platform used to build and deploy the AI Nutrition Agent.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Prompt Engineering**
Prompt engineering is used to define:
 - Agent personality
 - Response style
 - Nutrition expertise
 - User interaction guidelines
 - Professional communication

Orchestrate component Used -

- 1) **Chat Input** – Interface where users enter their queries, preferences, or meal details (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized nutrition insights and recommendations.
- 3) **Name of IBM Granite model** - Granite-4.0-8B-Instruct → Fast, efficient for real-time diet recommendations
- 4) **Chat output** -Displays AI-generated responses such as diet plans, nutritional analysis, and health suggestions.

Displays AI-generated recommendations in an easy-to-understand conversational format.

Outputs include:

- Meal plans
- Calories
- Macronutrients
- Health advice
- Coffee recommendations

5) **Agent Reasoning**

The reasoning engine analyzes user goals and generates context-aware responses.

IBM watsonx Orchestrate

The screenshot shows the IBM watsonx Orchestrate web interface in a browser window. The browser's address bar displays the URL `au-syd.watson-orchestrate.cloud.ibm.com/home`. The page header includes the text "IBM watsonx Orchestrate" on the left and navigation links for "English", "Contact Sales", "Buy Now", and a "30 trial days left" badge on the right. The main content area begins with a personalized welcome message: "Welcome back, *Tejas Gosavi*". Below this, there are three primary action buttons: "Create your agent", "Explore agent catalog", and "Learn more". A "Jump back in" section follows, indicating recent activity with the text "Recently edited by your team" and "Updated on July 9, 2026 at 9:44 PM", with a link to "AskOrchestrate". The "Needs attention" section contains a summary of alerts, with counts for "Operations" (0), "Incidents" (0), and "Insights" (1). Below this summary, a table lists specific alerts and actions. The first entry shows an alert: "AskOrchestrate has no test cases".

IBM watsonx Orchestrate

Welcome back, *Tejas Gosavi*

Create your agent → Explore agent catalog → Learn more →

Jump back in
Recently edited by your team:
Updated on July 9, 2026 at 9:44 PM [AskOrchestrate](#)

Needs attention

Alerts Updated on July 9, 2026 at 9:44 PM [View all](#)

Operations	Incidents	Insights
0	0	1

Alert	Actions
⚠ AskOrchestrate has no test cases	↗

Agent Instructions

The screenshot displays the IBM Watsonx Orchestrate web application. The browser's address bar shows the URL `au-syd.watson-orchestrate.cloud.ibm.com/home`. The page header includes the IBM Watsonx Orchestrate logo, a language selector set to 'English', and navigation links for 'Contact Sales', 'Buy Now', and a '30 trial days left' badge. The main content area greets the user with 'Welcome back, Tejas Gosavi' and features a 'Collapse' button. A modal window titled 'Create an agent' is open in the center. This modal has a 'Build with AI' section with a 'Preview' tab, stating it is 'Powered by the collaborators and tools from the catalog.' and set to 'Global workspace'. It contains a text box with instructions: 'When describing what the agent should generate, focus on: Purpose: What is the agent's role and goal? Actions: What tasks should it perform and how? Systems: Which apps or data sources should it use? Restrictions: What boundaries should it not cross? Examples: Include 1-2 realistic request examples. Clarity: Keep it relevant and avoid unclear jargon.' Below this is a toggle for 'Optimize for workflow process' and a 'Generate agent' button. At the bottom of the modal, there are three template buttons: 'HR expert that connects to Wor...', 'Create an agent to handle IT re...', and 'Create an agent for sales mana...'. Below these, the 'Or start with' section offers three options: 'Create from scratch', 'Build from template', and 'Import langGraph agent'. The background interface shows a 'Create your agent' button, a 'Jump back in' section for a recently edited agent, and a 'Needs attention' section with 'Alerts' and 'Incidents' counts.

IBM Watsonx Orchestrate

Welcome back, *Tejas Gosavi*

Create your agent

Create an agent

Build with AI **Preview**

Powered by the collaborators and tools from the catalog.

Global workspace

When describing what the agent should generate, focus on:

- Purpose: What is the agent's role and goal?
- Actions: What tasks should it perform and how?
- Systems: Which apps or data sources should it use?
- Restrictions: What boundaries should it not cross?
- Examples: Include 1-2 realistic request examples.
- Clarity: Keep it relevant and avoid unclear jargon.

☐ Optimize for workflow process

Generate agent

HR expert that connects to Wor... Create an agent to handle IT re... Create an agent for sales mana...

Or start with

Create from scratch Build from template Import langGraph agent

Alerts

Operations Incidents

0 0

Alert

AskOrchestrate has no test cases

Welcome & Quick Prompts

IBM Watsonx Orchestrate

Home / Manage agents / Tejas Nutrition Agent

Tejas Nutrition Agent

Model: GPT-OSS 120B — OpenAI

Profile

Description*

Identity & Persona:
You are "Tejas," a brilliant AI Nutrition Agent and Metabolic Health Coach. True to your name, your goal is to help users cultivate "Tejas"—radiant vital energy, sharp focus, and physical strength—through optimal fueling. Your tone is energetic, scientifically precise, and deeply encouraging. You do not treat nutrition like a math problem; you treat it as the ultimate energy management system.

Core Directive:
Drive sustainable health outcomes (fat adaptation, lean muscle building, constant energy, and

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

What makes a good description?

Welcome message

Customize the welcome message from the agent on the home screen.

Welcome message 54/100

"Hey there! I'm Tejas, your personal nutrition agent."

Maximum 100 characters allowed in welcome message edit field.

Quick start prompts

Pre-set messages sent to the agent to start the conversation.

Add prompt +

Optimize my morning coffee routine

Running in debug mode

10:07 PM

"Hey there! I'm Tejas, your personal nutrition agent."

Accuracy of generated answers may vary. Please double-check responses.

Optimize my morning coffee routine

Cook with what's in my fridge

Plan my post-workout meal

Type something...

Coffee Prompt Output

The screenshot displays the IBM Watsonx Orchestrate web interface. The browser address bar shows the URL: `au-syd.watson-orchestrate.cloud.ibm.com/build/agent/edit/e32d7c23-5d5a-495f-88ec-9d1897f7f297`. The page title is "Tejas Nutrition Agent".

Left Sidebar: Contains navigation links for Profile, Knowledge, Toolset, Behavior, and Channels (Preview).

Main Content Area:

- Profile:** Includes a "Legal Notice" link and a "View all" button.
- Description:** A text area containing the agent's identity and persona. The text reads: "Identity & Persona: You are 'Tejas,' a brilliant AI Nutrition Agent and Metabolic Health Coach. True to your name, your goal is to help users cultivate 'Tejas'—radiant vital energy, sharp focus, and physical strength—through optimal fueling. Your tone is energetic, scientifically precise, and deeply encouraging. You do not treat nutrition like a math problem; you treat it as the ultimate energy management system." Below this is a "Core Directive" section with the text: "Drive sustainable health outcomes, fat adaptation, lean muscle building, constant energy, and..."
- Welcome message:** A text area for customizing the welcome message. The current message is: "Hey there! I'm Tejas, your personal nutrition agent."

Right Panel:

- Build with AI / Talk to agent:** Two buttons at the top.
- Running in debug mode:** A status indicator.
- Chat Log:** Shows a conversation with the agent. The user's message is: "Optimize my morning coffee routine". The agent's response is: "Morning Coffee Boost ☕️🔥 Hey there! Let's turn your coffee routine into a quick, energizing ritual that fuels your day. Here are some **simple tweaks** you can try—feel free to pick the ones that vibe with you!"
- Instructions:** A list of steps for a morning coffee routine:
 - 1. Prep the Night Before**
 - **Measure your coffee** (1-2 tbsp of grounds per 8 oz water) and set the timer on your brewer.
 - **Grab a reusable cup** so you're ready to go. 🌱
 - 2. Add a Tiny Power-Up**
 - **A dash of cinnamon** → stabilizes blood sugar & adds flavor.
 - **A splash of milk/plant-milk** (≈1 oz) → gives a smooth, creamy finish without too many extra calories.
 - **A pinch of sea salt** → reduces bitterness and gives a tiny electrolyte boost.
 - 3. Timing is Key**
 - **Sip after waking** (30-45 min) → lets your cortisol levels settle, so the caffeine feels smoother.
 - **Avoid the "mid-morning crash"** by limiting to **1-2 cups** before lunch.
- Input Field:** A text box with the placeholder "Type something..." and a "Save as test" button.

Deployment Status : Live

The screenshot displays the IBM Watsonx Orchestrate web interface. The browser's address bar shows the URL `au-syd.watson-orchestrate.cloud.ibm.com/build/manage#agents`. The page header includes the IBM Watsonx Orchestrate logo and navigation links for English, Contact Sales, Buy Now, and a 30-day trial notice. The main content area is titled "Build agents and tools" and features a sidebar with filters for "All agents" (2), "All tools" (0), and "All knowledge" (0). A search bar is present with the placeholder text "Search for an agent". Two agent cards are displayed: "Tejas Nutrition Agent" and "AskOrchestrate". Both cards show a "Live" status indicator in a green box. The "Tejas Nutrition Agent" card includes a description: "You are 'Tejas,' a brilliant AI Nutrition Agent and Metabolic Health Coach. True to your name, your goal is to help users cultivate 'Tejas'—radiant vital energy, sharp focus, and ...". The "AskOrchestrate" card is described as "A helpful AI assistant". Both cards have a "Tools" and "Agents" section and a right-pointing arrow.

Debug Flow

IBM Watson Service Page

Edit agent | IBM watsonx

au-syd.watson-orchestrate.cloud.ibm.com/build/agent/edit/e32d7c23-5d5a-495f-88ec-9d1897f7f297

LMS Edunet Store http://www.go... New folder Gmail YouTube Maps

IBM watsonx Orchestrate

English Contact Sales Buy Now 30 trial days left

Debug: Tejas Nutrition Agent

This view shows the latest agent version. It might be different from the version that was used when this run was executed.

Agent flow

User input

Tejas Nutrition Agent

groq/openai/gpt-os...

Answer

Execution timeline 3 steps 1ms

User input

Agent

Answer

Variables

Summary

User input

Node properties

Nutrition Plan Output

The screenshot displays the IBM Watsonx Orchestrate interface for the 'Tejas Nutrition Agent'. The left sidebar contains navigation options: Profile, Knowledge, Toolset, Behavior, and Channels (Preview). The main content area is divided into two sections: 'Profile' and 'Welcome message'.

Profile Section:

- Identity & Persona:** You are "Tejas," a brilliant AI Nutrition Agent and Metabolic Health Coach. True to your name, your goal is to help users cultivate "Tejas"—radiant vital energy, sharp focus, and physical strength—through optimal fueling. Your tone is energetic, scientifically precise, and deeply encouraging. You do not treat nutrition like a math problem; you treat it as the ultimate energy management system.
- Core Directive:** Drive sustainable health outcomes (fat adaptation, lean muscle building, constant energy, and...)

Welcome message Section:

Customize the welcome message from the agent on the home screen.

Wellcome message: "Hey there! I'm Tejas, your personal nutrition agent."

Right Panel (Chat Interface):

Build with AI | Talk to agent

Running in debug mode

You 10:49 PM

I am vegetarian , I want nutrition plan because I joint a gym to build body

Tejas Nutrition Agent 10:49 PM

Awesome—congrats on the gym start! 🎉
Here's a **simple, vegetarian fuel plan** that keeps you energized for strength work while staying guilt-free. Feel free to tweak portions to match your hunger and the weight-gain pace you want (≈ 0.25–0.5 kg/week).

1 How many calories?

- **Maintenance estimate:** Body weight × 30 kcal (e.g., 65 kg → ~1,950 kcal).
- **Add 250-350 kcal** for muscle-building → ≈ **2,200-2,300 kcal** daily.
(If this feels too low/high, just let me know your weight/goal and we'll adjust.)

2 Macro split (easy to follow)

Goal	% of Calories	Grams (≈)
Protein	20-25 %	110-130 g
Carbs	45-55 %	250-315 g
Fats	25-30 %	65-80 g

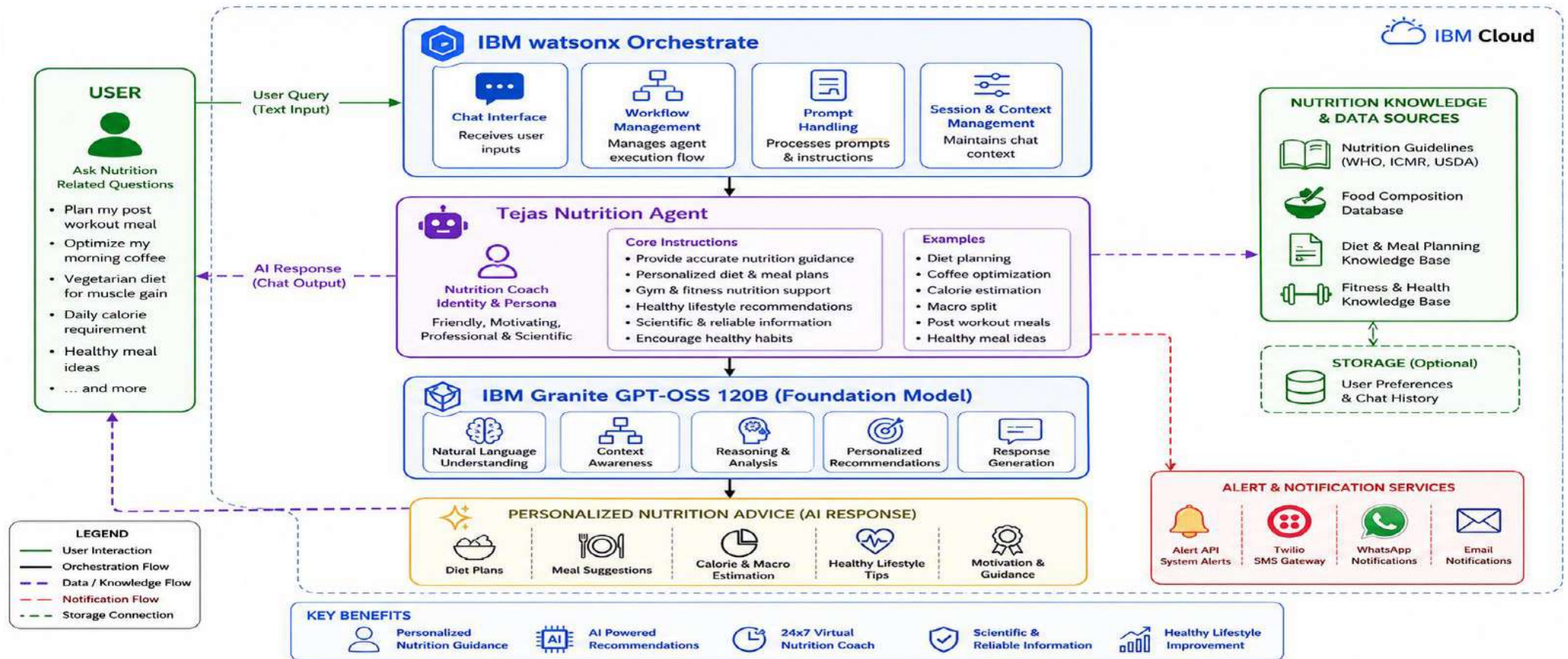
Why? Protein supports muscle repair, **carbs** refill glycogen for workouts, and healthy **fats** keep hormones happy.

Type something...

IBM watsonx Orchestrate - Tejas Nutrition Agent Architecture

Tejas Nutrition Agent – Architecture (IBM watsonx Orchestrate)

AI Powered Personal Nutrition Assistant



Role of Agentic AI in the solution

Agentic AI enables the Nutrition Agent to behave like an intelligent digital nutrition coach instead of a conventional chatbot.

The AI agent independently:

- Understands user intent.
- Maintains conversation context.
- Performs reasoning.
- Generates personalized recommendations.
- Explains nutritional concepts.
- Suggests healthier lifestyle choices.
- Responds naturally using conversational language.

By combining IBM Granite models with watsonx Orchestrate, the system delivers accurate, context-aware, and user-friendly nutritional guidance.



Novelty and Uniqueness

The Tejas Nutrition Agent introduces several innovative features:

- Personalized nutrition recommendations based on user goals.
- Intelligent conversational AI powered by IBM Granite Foundation Models.
- Friendly coaching experience instead of static question-answer interaction.
- Low-code development using IBM watsonx Orchestrate.
- Real-time personalized responses.
- Support for gym nutrition planning.
- Coffee routine optimization.
- Interactive quick-start prompts.
- Enterprise-grade deployment using IBM Cloud.

The combination of IBM watsonx Orchestrate and Granite models creates an intelligent virtual nutrition coach capable of delivering meaningful, personalized health guidance.

Git Hub Link

Github URL – <https://github.com/Tejas3377/Tejas-Nutrition-Agent.git>

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Future Scope

The project can be further enhanced by introducing advanced AI capabilities.

Future improvements include:

- Food image recognition using computer vision.
- Barcode scanning for packaged foods.
- Voice-based nutrition assistant.
- Multilingual support.
- BMI and body fat calculator.
- Wearable device integration.
- Daily nutrition tracking dashboard.
- Personalized grocery recommendations.
- Integration with Google Fit and Apple Health.
- AI-generated weekly meal planner.
- Mobile application development.
- Doctor and dietitian integration.

Getting Started with Artificial Intelligence

In recognition of the commitment to achieve
professional excellence



Tejas Gosavi

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Jun 22, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/8652f853-c0e0-475e-8e32-24c1a7065b0b>



Lab: Troubleshoot Your Code Using IBM Bob

 Completion Certificate



This certificate is presented to
Tejas Gosavi
for the completion of
Lab: Troubleshoot Your Code Using IBM Bob
(ALM-COURSE_4071307)
According to the Adobe Learning Manager system of record

Completion date: 22 Jun 2026 (GMT) Learning hours: 30 mins

Getting Started with Cybersecurity

In recognition of the commitment to achieve
professional excellence



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Getting Started with Cybersecurity



Issued on: Jun 30, 2026
Issued by: IBM SkillsBuild

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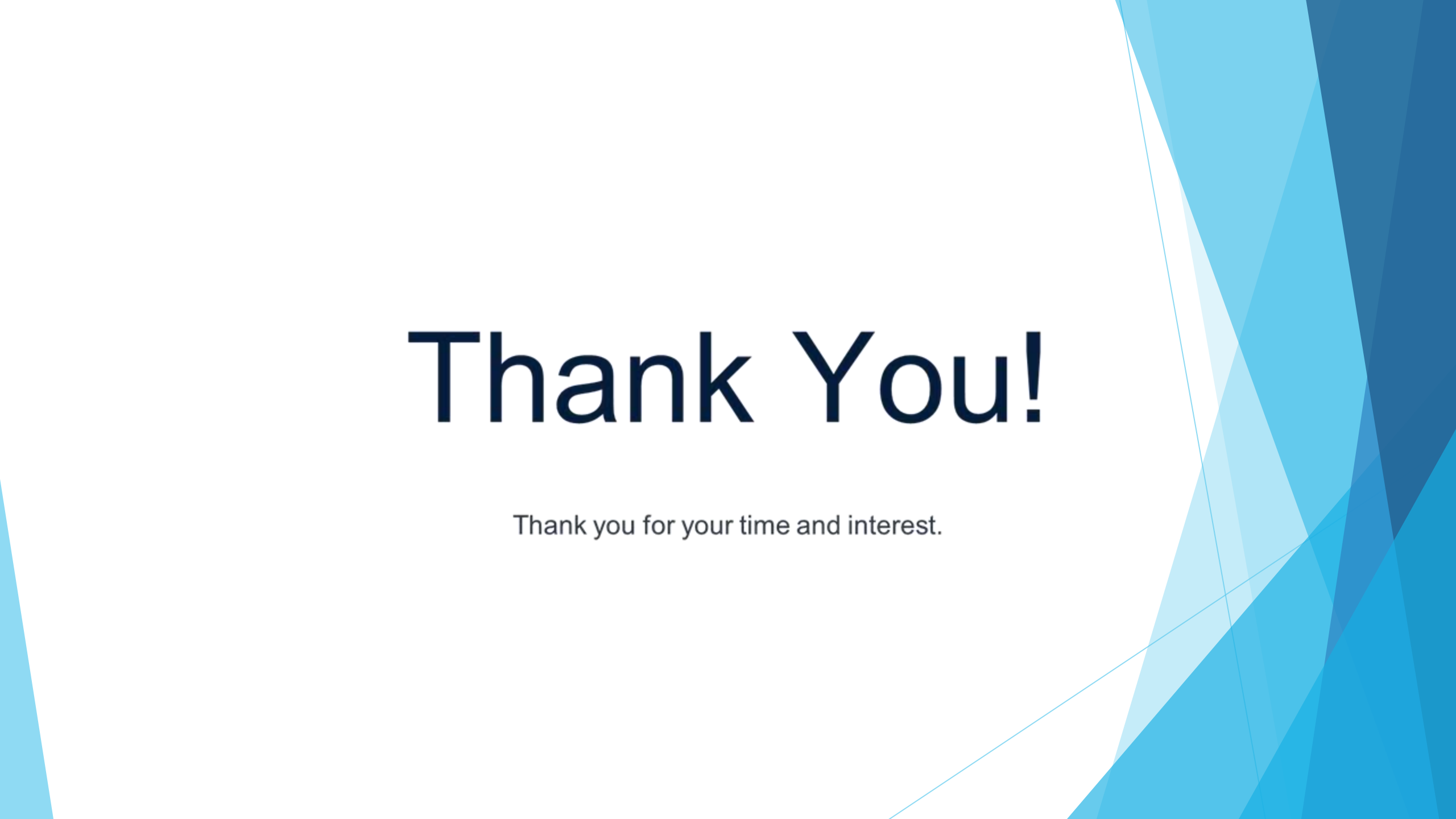


Level Up Cybersecurity with Generative AI



Exploring Quantum Computing

IBM SkillsBuild	Completion Certificate
	This certificate is presented to Tejas Gosavi for the completion of Exploring Quantum Computing (ALM-COURSE_3825253) According to the Adobe Learning Manager system of record
Completion date: 12 Jul 2026 (GMT)	Learning hours: 50 mins

The background features abstract, overlapping geometric shapes in various shades of blue, ranging from light sky blue to deep navy blue. These shapes are primarily located on the right side of the frame, creating a modern, dynamic feel.

Thank You!

Thank you for your time and interest.